Native GPU-Rasterized PDF Pipeline

Section 1: Abstract and System Foundations

The StarPDF GPU rasterizer provides hardware-accelerated rendering directly from the pure-Rust StarPDF Render IR. Prior implementations relied on CPU rasterization via tiny-skia, incurring significant CPU overhead on complex vector shapes and dense text layouts. Phase 7 and Phase 8 introduce native TrueType/OpenType glyph outline tessellation and strict sequential z-order compositing without a global CPU overlay.

[Figure 1: Dual-Phase Command Execution Architecture]

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Phase 1: Parse Render IR -> Tessellate Lyon paths & Glyphs -> Pre-allocate Buffers

Phase 2: Begin wgpu::RenderPass -> Sequential GpuDrawCmd Execution -> Submit

Clipping: Hardware Scissor Rectangles + Scoped CPU Transparency Fallbacks

Performance: >15x speedup over CPU on text-heavy mixed pages
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The compilation phase maps PDF content streams into typed RenderItem representations. Text runs contain codepoints, matrix transformations, and font references. Vector paths contain cubic and quadratic beziers.

All vertex coordinates undergo scale and translation transformations directly mapped to device pixels. Hardware scissor rectangles isolate clipping regions without requiring multi-pass stencil buffers.

Section 2: Multi-Column Memory and Layout Profiling

Benchmark Target	CPU tiny-skia (ms)	GPU Render (ms)	Pass Speedup
`text-multicol-table.pdf`	49.46 ms	2.22 ms	22.28x
`text-multispan-heading.pdf`	48.14 ms	3.00 ms	16.05x
`scanned-test.pdf`	323.22 ms	1.96 ms	164.72x
`smartpdf-adobe-like-form.pdf`	28.35 ms	2.56 ms	11.09x

The table above underscores the empirical performance advantages observed on Apple Silicon Metal targets. GPU execution demonstrates near constant-time rasterization across complex glyph metrics. Memory allocations remain bounded under 250 MB even during rapid zoom sweeps and tab transitions.

Section 3: Qualification Matrix & Conclusions

Real-world qualification requires robust handling of non-standard fonts, broken xref tables, and legacy PDF operators. By preserving fail-safe CPU fallbacks for unsupported shading models while delegating 100% of standard vector paths, text glyphs, and images to the GPU, StarPDF delivers production stability with zero silent degradation.

VERIFIED BY: ToolRakyat Desktop Engineering Team

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